

Python Programming Assignment

Topic: Decision Making Statements (if-else & if-elif-else)

Course: Python Fundamentals

Total Questions: 33

Level: Beginner to Intermediate

Part 1: if-else Conditional Statements (Questions 1 – 25)

1. Smallest of Two Numbers

Given two integers, write a program to find the smallest number using an `if-else` statement.

INPUT FORMAT: Line 1: Integer `A` | Line 2: Integer `B`

OUTPUT FORMAT: Print the smallest number.

CONSTRAINTS: $-10^9 \leq A, B \leq 10^9$

SAMPLE INPUT	SAMPLE OUTPUT
25 15	15

2. Largest of Two Numbers

Given two integers, find the largest number using an `if-else` statement.

INPUT FORMAT: Two integers `A` and `B` .

OUTPUT FORMAT: Print the largest number.

SAMPLE INPUT	SAMPLE OUTPUT
25 40	40

3. Absolute Value

Given an integer, find its absolute value. If the number is negative, convert it into its positive value. Otherwise, print the number as it is. **Do not use the built-in `abs()` function.**

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print the absolute value of `N` .

SAMPLE INPUT	SAMPLE OUTPUT
-25	25

4. Even or Odd

Given an integer, determine whether the number is even or odd.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Even` if the number is even. Otherwise, print `Odd` .

SAMPLE INPUT	SAMPLE OUTPUT
28	Even

5. Multiple of 5

Given an integer, determine whether it is a multiple of 5.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Multiple of 5` if divisible by 5. Otherwise, print `Not a Multiple of 5` .

SAMPLE INPUT	SAMPLE OUTPUT
35	Multiple of 5

6. Multiple of 10

Given an integer, determine whether it is a multiple of 10.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Multiple of 10` or `Not a Multiple of 10` .

SAMPLE INPUT	SAMPLE OUTPUT
120	Multiple of 10

7. Two-Digit Number

Given an integer, check whether it is a two-digit number.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Two Digit Number` if it contains exactly two digits. Otherwise, print `Not a Two Digit Number` .

SAMPLE INPUT	SAMPLE OUTPUT
45	Two Digit Number

8. Three-Digit Number

Given an integer, check whether it is a three-digit number.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Three Digit Number` or `Not a Three Digit Number` .

SAMPLE INPUT	SAMPLE OUTPUT
125	Three Digit Number

9. Ends with Zero

Given an integer, determine whether the number ends with zero.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Ends with Zero` or `Does Not End with Zero` .

SAMPLE INPUT	SAMPLE OUTPUT
450	Ends with Zero

10. Square Comparison

Given an integer, calculate its square. Check whether the square is greater than 50 or less than or equal to 50.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Above 50` if the square is > 50 , otherwise print `50 or Below` .

SAMPLE INPUT	SAMPLE OUTPUT
8	Above 50

11. Difference Check

Given two integers, subtract the second number from the first number. Check whether the resulting difference is zero or not.

INPUT FORMAT: Two integers `A` and `B` .

OUTPUT FORMAT: Print `Difference is Zero` if 0. Otherwise, print `Difference is Not Zero` .

SAMPLE INPUT	SAMPLE OUTPUT
25 25	Difference is Zero

12. Student Pass or Fail

A student passes an examination if the Computer Science mark is **50 or above**. Given the student's mark, determine whether the student has passed or failed.

INPUT FORMAT: A single integer `marks` .

OUTPUT FORMAT: Print `Pass` if $\text{mark} \geq 50$. Otherwise, print `Fail` .

SAMPLE INPUT	SAMPLE OUTPUT
72	Pass

13. Divisible by 10

Given an integer, determine whether it is divisible by 10.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Divisible by 10` or `Not Divisible by 10` .

SAMPLE INPUT	SAMPLE OUTPUT
250	Divisible by 10

14. Biggest Digit of a Two-Digit Number

Given a two-digit integer, find the biggest digit.

INPUT FORMAT: A two-digit integer `N` .

OUTPUT FORMAT: Print the biggest digit.

SAMPLE INPUT	SAMPLE OUTPUT
47	7

15. Exam Difficulty

An examination system receives a choice from the user. If the choice is `1` , the exam is easy. For any other choice, the exam is difficult.

INPUT FORMAT: A single integer `choice` .

OUTPUT FORMAT: Print `The exam will be easy` if choice is 1, otherwise print `The exam will be difficult` .

SAMPLE INPUT	SAMPLE OUTPUT
1	The exam will be easy

16. Permission to Play

A student enters a value. If the value is `1`, the student is allowed to play. Otherwise, they cannot play.

INPUT FORMAT: A single integer `value`.

OUTPUT FORMAT: Print `You can go out and play` if 1, otherwise print `You cannot go out and play`.

SAMPLE INPUT	SAMPLE OUTPUT
1	You can go out and play

17. Square or Rectangle

Given length and breadth, determine whether the shape is a square or rectangle. Equal dimensions form a square.

INPUT FORMAT: Line 1: Length | Line 2: Breadth

OUTPUT FORMAT: Print `Square` or `Rectangle`.

SAMPLE INPUT	SAMPLE OUTPUT
10 10	Square

18. Uppercase ASCII Value

Given an ASCII value, check if it represents an uppercase English alphabet (Range: 65 to 90).

INPUT FORMAT: A single integer `N`.

OUTPUT FORMAT: Print `Uppercase Alphabet` or `Not an Uppercase Alphabet`.

SAMPLE INPUT	SAMPLE OUTPUT
70	Uppercase Alphabet

19. Lowercase ASCII Value

Given an ASCII value, check if it represents a lowercase English alphabet (Range: 97 to 122).

INPUT FORMAT: A single integer `N`.

OUTPUT FORMAT: Print `Lowercase Alphabet` or `Not a Lowercase Alphabet`.

SAMPLE INPUT	SAMPLE OUTPUT
105	Lowercase Alphabet

20. Numeric Character ASCII Value

Given an ASCII value, check if it represents a numeric character ('0'-'9', Range: 48 to 57).

INPUT FORMAT: A single integer `N`.

OUTPUT FORMAT: Print `Numeric Character` or `Not a Numeric Character`.

SAMPLE INPUT	SAMPLE OUTPUT
53	Numeric Character

21. Multiple of 5 and 3

Given an integer, determine whether it is a multiple of both 5 and 3.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Multiple of 5 and 3` or `Not a Multiple of 5 and 3` .

SAMPLE INPUT	SAMPLE OUTPUT
30	Multiple of 5 and 3

22. Three-Digit Multiple of 10

Given an integer, check if it is both a three-digit number and a multiple of 10.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Valid Number` if both conditions are met. Otherwise, print `Invalid Number` .

SAMPLE INPUT	SAMPLE OUTPUT
250	Valid Number

23. Three-Digit Multiple of 2, 5 and 10

Given an integer, check whether it is a three-digit number AND simultaneously a multiple of 2, 5, and 10.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Valid Number` or `Invalid Number` .

SAMPLE INPUT	SAMPLE OUTPUT
250	Valid Number

24. Product or Sum

Given two integers: If both numbers are even, calculate their product. Otherwise, calculate their sum.

INPUT FORMAT: Two integers `A` and `B` .

OUTPUT FORMAT: Print the calculated result.

SAMPLE INPUT	SAMPLE OUTPUT
4 6	24

25. Buzz Number

A number is called a **Buzz Number** if it ends with 7 OR is divisible by 7. Check if a given integer is a Buzz Number.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Buzz Number` or `Not a Buzz Number` .

SAMPLE INPUT	SAMPLE OUTPUT
1007	Buzz Number

Part 2: if-elif-else Conditional Statements (Questions 1 – 8)

1. Largest of Three Numbers

Given three integers, find the largest number using `if-elif-else` .

INPUT FORMAT: Three integers `A` , `B` , and `C` .

OUTPUT FORMAT: Print the largest number.

SAMPLE INPUT	SAMPLE OUTPUT
25 78 45	78

2. Smallest of Three Numbers

Given three integers, find the smallest number using `if-elif-else` .

INPUT FORMAT: Three integers `A` , `B` , and `C` .

OUTPUT FORMAT: Print the smallest number.

SAMPLE INPUT	SAMPLE OUTPUT
25 78 12	12

3. Positive, Negative or Zero

Given an integer, print `Positive` if > 0 , `Negative` if < 0 , and `Zero` if equal to 0.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Positive` , `Negative` , or `Zero` .

SAMPLE INPUT	SAMPLE OUTPUT
-15	Negative

4. Library Fine

A library charges late fines per book based on the schedule below:

Late Days	Fine per Day
First 5 days (1–5)	40 paisa
6–10 days	65 paisa
Above 10 days	80 paisa

INPUT FORMAT: A single integer `days` .

OUTPUT FORMAT: Print total fine in paisa.

SAMPLE INPUT	SAMPLE OUTPUT
7	455

5. Basic Calculator

Create a basic calculator accepting two numbers and an operator (`+` , `-` , `x` , `/`). Print the result.

INPUT FORMAT: Line 1: Number A | Line 2: Number B | Line 3: Operator

OUTPUT FORMAT: Print result of the selected operation.

SAMPLE INPUT	SAMPLE OUTPUT
20 5 +	25

6. Multiple of 5, 3 and 7

Given an integer, determine whether it is a multiple of 5, 3, and 7 simultaneously.

INPUT FORMAT: A single integer `N` .

OUTPUT FORMAT: Print `Multiple of 5, 3 and 7` or `Not a Multiple` .

SAMPLE INPUT	SAMPLE OUTPUT
105	Multiple of 5, 3 and 7

7. Parcel Booking Charges Complete Tariff

A courier company calculates delivery charges based on weight (in kg) and booking type (`O` for Ordinary, `E` for Express):

Weight Slab (in kg)	Ordinary ('O') Rate	Express ('E') Rate
Up to 5 kg	₹ 50 / kg	₹ 80 / kg
Next 5 kg (6–10 kg)	₹ 40 / kg	₹ 70 / kg
Above 10 kg	₹ 30 / kg	₹ 50 / kg

INPUT FORMAT: Line 1: Weight (kg) | Line 2: Booking Type (`O` or `E`)

OUTPUT FORMAT: Print total calculated charge.

SAMPLE INPUT	SAMPLE OUTPUT
8 E	610
4 O	200

8. Laptop Discount Calculator Complete Slabs

An electronics store offers tiered discounts on laptops based on original purchase price:

Laptop Price Range (in ₹)	Discount Rate (%)
Up to ₹ 25,000	5 %
₹ 25,001 to ₹ 50,000	10 %
₹ 50,001 to ₹ 1,00,000	15 %
Above ₹ 1,00,000	20 %

INPUT FORMAT: A single numeric value representing laptop price.

OUTPUT FORMAT: Display in 3 lines:

```
Price of laptop : <price>
Discount : <discount>
Total Price : <total_price>
```

SAMPLE INPUT	SAMPLE OUTPUT
50000	Price of laptop : 50000 Discount : 5000.0 Total Price : 45000.0
65000	Price of laptop : 65000 Discount : 9750.0 Total Price : 55250.0